Lecture 8 (03/06/21)

The Complete Solution of $Ax = b$

- Elimination is one step towards the solution

Example: (Equations 3, Unknowns 4)

$$x_1 + 2x_2 + 2x_3 + 2x_4 = b_1$$
$$2x_1 + 4x_2 + 6x_3 + 8x_4 = b_2$$
$$3x_1 + 6x_2 + 8x_3 + 10x_4 = b_3$$

Matrix form

$$\left[\begin{array}{cccc:c} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{array}\right]$$

can we solve x for any values of $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$? . No,

No. $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ must be in the columnspace of A,

Only then it will solve the system of linear equations.

$R_2 = R_2 - 2.R_1$

$R_3 = R_3 - 3.R_1$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 - 3b_1 \end{array}\right]$$

$R_3 = R_3 - R_2$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{b_3 - b_2 - b_1} \end{array}\right]$$

Solvability condition here,

$$b_3 - b_2 - b_1 = 0$$

$$\Rightarrow \boxed{b_3 = b_1 + b_2}$$

$Ax = b$ will be solvable when b is in the columnspace of A.

→ If combination of rows of A gives us 0, same combinations of corresponding bs will give us 0 too. (simply → same value)

2 Steps to find complete solution

<u>Step 1</u>: Find Xparticular (x_p)

- Set all free variables to 0.

$$\left[\begin{array}{cccc:c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 - b_1 \end{array}\right]$$

Solve for real values
$b_3 = b_1 + b_2$ (satisfied)

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$$

$$\left[\begin{array}{cccc:c} 1 & 2 & 2 & 2 & 1 \\ 0 & 0 & 2 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \rightarrow \begin{aligned} x_1 + 2x_2 + 2x_3 + 2x_4 &= 1 \\ 2x_3 + 4x_4 &= 3 \end{aligned}$$

Free columns
$\underline{\underline{x_2, x_4}}$
free variables

$$\begin{aligned} x_1 + 2x_3 &= 1 \\ 2x_3 &= 3 \end{aligned}$$

$$x_3 = 3/2$$

$$x_1 = -2$$

$$x_p = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}$$

Step 2 : $x_{complete} = x_{particular} + x_n = x_p + x_n$

x_n = nullspace of A.

Why ?

$$\begin{array}{r} Ax_p = b \\ Ax_n = 0 \\ \hline A(x_p + x_n) = b \end{array}$$

$\therefore \boxed{Ax_c = b}$

Not subspace but a solutionspace.

$$x_n = c\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d\begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$ {prev. lecture}

$\therefore x_{complete} = x_p + x_n$

$$= \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + \left\{ c\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d\begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

Check ? put $c, d \in \mathbb{R}$ and find $Ax_c = b$.

What is the difference between x_p and x_n?

x_p is sl^n when b has non-zero value for $Ax = b$.

x_n is " " " " zero " " $Ax = 0$.

while solving for x_p, all free variables must be set to 0.

For $x_{special}$ → all free variables must be set to 0 except one / "One of the free variables must be set to 1"

$$x_c = x_p + x_n$$

$$= \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + \left\{ c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

What is the number of sl^ns of x_c here?

Infinite $c, d \in \mathbb{R}$

There might be a case where the R.H.S b doesn't fall inside the columnspace, there might be no sl^n then

An $m \times n$ matrix of rank 'r' (pivots)

What's the relation between r & m?

$$\boxed{r \leq m}$$

What's the relation between r & n?

$$\boxed{r \leq n}$$

$$\therefore r = \min(m, n)$$

Case 1 : Full column rank, $r = n$

$$A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \\ 6 & 1 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} \boxed{\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}} \\ \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} \end{bmatrix} = \begin{bmatrix} I \\ 0 \end{bmatrix}$$

* independent columns, no free variables.
* no free variables → can't set 1 to any columns while solving for nullspace (no free columns)

// So, we will only get 1 soln for nullspace, $N(A) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

// "Nullspace of matrix with independent columns is zero vector"

So, in this case,

for solving $Ax = b$, $x_c = x_p$ as $x_n = 0$

$\therefore Ax_p = b$ [Unique Solution if exists]

[there might be a case where b isn't in the columnspace of A. Then no solution]

Case 1 Summary:

If a matrix is full column rank

1. All columns of A are pivot columns

2. There are no free variables ; no special sol^n

So, the nullspace is zero vector.

3. If $Ax = b$ has a solution (it might not), then its the only (unique) sol^n.

[b must be in columnspace of A]

Case 2: Full row rank, $r = m$ $(m \leq n)$

(every row has a pivot, rows linearly independent)

$$x + y + z = 3$$
$$x + 2y - z = 4$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & -1 & 4 \end{array}\right]$$

//

$$= \left[\begin{array}{cc|cc} 1 & 0 & 3 & 2 \\ 0 & 1 & -2 & 1 \end{array}\right] = \left[\begin{array}{c|c} I & F \end{array}\right]$$

I — free column

(row reduced form)

//

$$x_p = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \qquad x_n = c \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

All rows has a pivot, nullspace not empty.

$Ax = b$ has a sl^n for all R.H.S b (no solvability cnd)

independent row = 1 pivot per row

entries of b doesn't have to depend on other elements

Case 2 summary

If a matrix is full row rank

1. All rows have pivots
2. $Ax = b$ has a solution for all r.h.s b.
3. The columnspace is R^m

   (m pivots for m rows, columnspace can span the whole of the rowspace)

4. Nullspace of A is non-empty
5. There are $n - r = n - m$ $(r = m)$

   Special solutions in $N(A)$.

6. No. of solution infinite. $x_c = x_p + \boxed{x_n}$

   Always get a sl^n for $Ax = b$, and $x_n \neq$ Empty

Case 3 : Full rank matrix $r = m = n$ [square matrix]

If A is invertible, nullspace is 0, independent col^{n}s.

$$\begin{bmatrix} x & x & x \\ x & x & x \\ x & x & x \end{bmatrix} \longrightarrow \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Row reduced form}} \quad \therefore N(A) \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$I$$

We can reach out to any point in respective dimension.

Columnspace spans whole vectorspace. $\rightarrow I$

Always get a sl^n for any combination of b,

Case 3 Summary

$\rightarrow$ One unique solution for every b.

Summary Table

	Full rank $r=m=n$	full column rank $r=n<m$	Full row rank $r=m<n$	$r<m, r<n$
Row Reduced Echelon form R (after elimination)	I	$\begin{bmatrix} I \\ 0 \end{bmatrix}$	$\begin{bmatrix} I & F \end{bmatrix}$	$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
No. of solutions for $Ax=b$	Exacty 1	0 if b not in col. space. 1 if b in col. space	Infinitely many	$0 \to$ no sln $\infty \to N(A)$ not empty

Example:

find all solutions depending on b_1, b_2, b_3

System of Eq^{n}s,

$$x - 2y - 2z = b_1$$
$$2x - 5y - 4z = b_2$$
$$4x - 9y - 8z = b_3$$

Matrix

$$\left[\begin{array}{ccc:c} 1 & -2 & -2 & b_1 \\ 2 & -5 & -4 & b_2 \\ 4 & -9 & -8 & b_3 \end{array}\right]$$

$$\rightarrow \underbrace{\left[\begin{array}{ccc:c} 1 & -2 & -2 & b_1 \\ 0 & -1 & 0 & -2b_1 + b_2 \\ 0 & 0 & 0 & -2b_1 - b_2 + b_3 \end{array}\right]}_{U} \rightarrow \underbrace{\left[\begin{array}{ccc:c} 1 & 0 & -2 & 5b_1 - 2b_2 \\ 0 & -1 & 0 & 2b_1 - b_2 \\ 0 & 0 & 0 & -2b_1 - b_2 + b_3 \end{array}\right]}_{R}$$

$$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

Solvability cond^n, $-2b_1 - b_2 + b_3 = 0$ [from U,

* If Not fulfilled, 0 sl^{n}s.
* If fulfilled, for $Ax = b$ [b is solvable]

Special sl^n, $Ax = 0$ - [Free vl^n = 1]

From R,

$$x_1 - 2x_3 = 0 \qquad x_3 = 1,$$
$$x_2 = 0 \qquad x_2 = 0$$
$$x_1 = 2$$

$$\therefore x_s = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$x_{complete} = x_p + x_n$$

$$= x_p + c \cdot x_s$$

$$= \begin{bmatrix} 5b_1 - 2b_2 \\ 2b_1 - b_2 \\ 0 \end{bmatrix} + c \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Conditions on x_p,

$$x_1 - 2x_3 = 5b_1 - 2b_2 \qquad \text{free var} = 0$$
$$x_2 = 2b_1 - b_2 \qquad \therefore x_3 = 0$$

$$\therefore x_p = \begin{bmatrix} 5b_1 - 2b_2 \\ 2b_1 - b_2 \\ 0 \end{bmatrix}$$